ERD Design Rationale

MoMo SMS Data Processing System, Design Justification

The ERD database schema was designed after the analysis of all 1,691 SMS messages in the provided “modified_sms_v2.xml”. This is to ensure that every decision made is founded from real data rather than assumptions.

Our schema contains six entities. ‘**Users’** stores every individual referenced in the data provided. This includes the account owner, personal contacts, merchants, agents, and the MTN System user for automated messages (airtime and data bundles).
‘Accounts’ were added after we came to the realization that the account owner holds two distinct numbers: a MoMo wallet (36521838) and what seems to be a linked I&M Bank Account (20077201001). A single entry in users couldn’t accommodate the 1 to many relationship between user and accounts.
‘Transaction_Category’ normalizes the 11 different SMS transaction patterns that include but are not limited to INCOMING_TRANSFER, MERCHANT_PAYMENT and CASH_WITHDRAWAL. Into a lookup table, each carrying a direction attribute (CREDIT, DEBIT) so the dashboard can compute balance movement.
‘Transactions’ entity is the central entity in our system. It records the financial details of each parsed SMS.
‘Transaction_Participants’ resolves the required many to many relationship between Users and Transaction by storing one row per individual per transaction with roles that include SENDER, RECEIVER, or AGENT.
Lastly, **‘System_Logs’** provides a full audit trail for the ETL (Extract, Transform, Load) pipeline. It’s automatic entries written by the database triggers whenever a transaction is recorded as either FAILED or REVERSED.

In addition to the main entities, within our data we specified specific fields as **‘Nullable’**. Meaning that this piece of information is not a must to be included in the entry. These attributes include: phone_number, balance_after, account_id, and merchant_code. This allows us to ensure we deal with all the gaps in the observed SMS data rather than having any design oversights. All of the monetary values have been marked as **DECIMAL** to prevent floating-point errors, and fixed-value columns use ENUM to enforce specific data.

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